11. An equation of the form $f_1(x)g_1(y)dx + f_2(x)g_2(y)dy = 0$ is called an equation with variable separable or simply a separable equation.
12. A function $f(x, y)$ is said to be a **homogeneous** function of degree n in the variables x and y if, $f(tx, ty) = t^n f(x, y)$ for some $n \in \mathbb{R}$ for all suitably restricted x, y and t. This is known as **Euler's homogeneity.**
13. If $f(x, y)$ is a homogeneous function of degree zero, then there exists a function g such that $f(x, y)$ is always expressed in the form $g\left(\frac{y}{x}\right)$.
14. An ordinary differential equation is said to be in homogeneous form, if the differential equation is written as $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$.
15. The differential equation $M(x, y)dx + N(x, y)dy = 0$ [in differential form] is said to be **homogeneous** if M and N are **homogeneous functions of the same degree**.
16. A **first order differential equation** of the form $\frac{dy}{dx} + Py = Q$.
    where P and Q are functions of x only. Here no product of y and its derivative $\frac{dy}{dx}$ occurs and the dependent variable y and its derivative with respect to independent variable x occur only in the first degree.
    The solution of the given differential equation (1) is given by $ye^{\int Pdx} = \int Qe^{\int Pdx}dx + C$.
    Here $e^{\int Pdx}$ is known as the integrating factor (I.F.)
17. A first order differential equation of the form $\frac{dx}{dy} + Px = Q$, where P and Q are functions of y only. Here no product of x and its derivative $\frac{dx}{dy}$ occurs and the dependent variable x and its derivative with respect to independent variable y occur only in the first degree.In this case, the solution is given by $xe^{\int Pdy} = \int Qe^{\int Pdy}dy + C$.
18. If x denotes the amount of the quantity present at time t, then the instantaneous rate at which the quantity changes at time t is $\frac{dx}{dt}$.
    This leads to a differential equation of the form $\frac{dx}{dt} = f(x, t)$.

ICT CORNER

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